CSE 2046/2246 Quiz

28. 04. 2023

1 Sessions (30 minutes)

Cheat Sheets are not allowed!

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Take it easy!

Q1. (20 pts) For each of the following, indicate the class Θ(g(n)) the function belongs to. Use the simplest g(n) possible.

(a) $\ln(3^{2n}) + \ln(n^{3e})$

(b) $5^{n/2} + n^3 2^n$

(c) $(2n)! + 4^n$

(d) $2^{n+1} \cdot \log_{10} \sqrt{n} + n^{10}$

Q2. (40 pts) How many multiplications (in terms of n) will be performed in the following function? Write a recurrence relation and solve it by backward substitution.

```
function f(n)
  if n <= 1 return 4
  else return f(n-1)*f(n-1)*f(n-1)*f(n-1);
```

Q3. (40 pts) A, B and C are three integer arrays of size m, n and t respectively. You are asked to print all the integers that exist in both A and C, and does not exist in B. Write a pseudocode for a brute-force algorithm and give its time complexity (in worst and best case).

Q1. a) $\ln(3^{2n}) + \ln(n^{3e}) = \underbrace{2n \ln 3}_{\Theta(n)} + \underbrace{3e \ln n}_{\Theta(\log n)} \in \Theta(n)$

b) $5^{n/2} + n^3 2^n = (\sqrt{5})^n + n^3 \cdot 2^n \in \Theta((\sqrt{5})^n)$ $\lim_{n\to\infty} \frac{(\sqrt{5})^n}{n^3 \cdot 2^n} = \infty$

c) $(2n)! + 4^n \in \Theta((2n)!)$ factorial has higher asymptotic growth. Note that $(2n)! \notin \Theta(n!)$

d) $2^{n+1} \cdot \log_{10} \sqrt{n} + n^{10} = \underbrace{2 \cdot 2^n \cdot \frac{1}{2} \log n}_{\Theta(\log n \cdot 2^n)} + \underbrace{n^{10}}_{\Theta(n^{10})} \in \Theta(\log n \cdot 2^n)$

Q2. $M(0) = M(1) = 0$

$M(n) = 4M(n-1) + 3$

$= 4(4M(n-2)+3)+3 = 4^2 M(n-2) + 4 \cdot 3 + 3$

$= 4^2(4M(n-3)+3) + 4 \cdot 3 + 3 = 4^3 M(n-3) + 4^2 \cdot 3 + 4 \cdot 3 + 3$

$\vdots$

$= 4^{n-1} M(1) + 4^{n-2} \cdot 3 + 4^{n-3} \cdot 3 + \ldots + 4 \cdot 3 + 3$

$= 0 + 3(1 + 4 + 4^2 + \ldots 4^{n-2}) = 3 \cdot \frac{4^{n-1} - 1}{4-1} = 4^{n-1} - 1 \in \Theta(4^n)$.

Q3.

```
for i = 1 to m
    for j = 1 to t
        if (A[i] = C[j]){
            exist ← 0
            for k = 1 to n
                if (A[i] = B[k])
                    exist ← 1
            if exist = 0
                print A[i]
        }
```

In the worst case, all the elements of A are the same as all the elements of C, and the first if statement is always true. In this case the basic operation is the comparison "A[i]=B[k]", and it is executed $\sum_{i=1}^{m} \sum_{j=1}^{t} \sum_{k=1}^{n} 1 = m.t.n$ times.

In the best case the first if statement never holds, and the number of comparisons would be m.t

So, worst case time complexity is $\Theta(mnt)$

Best case time complexity is $\Theta(mt)$